

AMTCH → Srishekhar de

- ODE: One independent variable
- PDE: more than one indep variable

* CHARACTERISTICS OF PDES

1) Order of PDE - Highest derivative in Eqⁿ
Eg > $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} \rightarrow \text{order 2}$

$$\frac{\partial u}{\partial t} = \frac{\partial u}{\partial x} + \frac{\partial^2 u}{\partial x^2} + \frac{\partial^3 u}{\partial x^3} \rightarrow \text{order 3}$$

2) Linear & Non Linear PDE:-

- a) Product of Dependent variable and its derivative and:
- b) Power of dependent variable and derivative is not 1,
then the PDE is non linear.

Example: $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ $\frac{\partial u}{\partial t} = u \frac{\partial^2 u}{\partial x^2}$ $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + u^2$

Linear

3) Homogeneous / Non Homogeneous PDE:- If equation has a term which is constant / function of independent variable, equation is non homogeneous

Eg > $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + x$ $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + 4$ $\frac{\partial^2 u}{\partial x^2} + x t = \frac{\partial u}{\partial t}$

* BOUNDARY CONDITIONS

↙ ↘
OPEN BOUNDARY CLOSED BOUNDARY

Normal, most basic



1) DIRECT MIXED B.C. :- Value of the dependent variable is specified on the boundary
At $x=1$; $T=T_0$; $x=0$, $T=0$

Derivative at boundary is 0 or constant

2) NEUMANN B.C. :- Normal Derivative of dependent variable is specified.
At $x=1$, $\frac{\partial T}{\partial x} = 0$; $x=0$, $\frac{\partial T}{\partial x} = -q/k$

weighted combination of dirichlet & neuman

3) ROBIN MIXED B.C. :- Dependent variable and its derivative are connected by a simple algebraic eqⁿ on boundary.
At $x=L$, $-k \frac{\partial T}{\partial x} = h(T-T_0)$

4) CAUCHY BC :- Two conditions are specified on the same boundary.

At $t=0$, $u = u_0$ & $\frac{\partial u}{\partial t} = u_{02}$

5) PHYSICAL B.C :- For flow through a pipe : At $x=R$, $u=0$
Not given explicitly $\rightarrow x=0$, $u = \text{const}$ or $\frac{\partial u}{\partial x} = 0$

* TYPES OF PDES :

Generalized form of PDEs of Order 2:-

$$\sum_{i=1}^3 \sum_{j=1}^3 a_{ij} \frac{\partial^2 u}{\partial x_i \partial x_j} = R(x_1, x_2, x_3, \frac{\partial u}{\partial x_1}, \frac{\partial u}{\partial x_2}, \frac{\partial u}{\partial x_3})$$



$$a_{11} \frac{\partial^2 u}{\partial x_1^2} + a_{12} \frac{\partial^2 u}{\partial x_1 \partial x_2} + a_{13} \frac{\partial^2 u}{\partial x_1 \partial x_3}$$

$$+ a_{21} \frac{\partial^2 u}{\partial x_2 \partial x_1} + a_{22} \frac{\partial^2 u}{\partial x_2^2} + a_{23} \frac{\partial^2 u}{\partial x_2 \partial x_3}$$

$$+ a_{31} \frac{\partial^2 u}{\partial x_3 \partial x_1} + a_{32} \frac{\partial^2 u}{\partial x_3 \partial x_2} + a_{33} \frac{\partial^2 u}{\partial x_3^2}$$

$$\partial x_3 \partial x_1$$

$$\partial x_3 \partial x_2$$

$$\partial x_3^2$$

Let $A = \text{Coeff matrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$

- on evaluating eigenvalues of A :-

- If eigenvalues are of same sign $\Rightarrow$ Elliptical PDE
- If atleast one eigenvalue is 0 $\Rightarrow$ Parabolic PDE
- If eigenvalues are of mixed sign $\Rightarrow$ Hyperbolic PDE

Ex > $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0 \Rightarrow A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ EV = (1, 1, 1) Elliptical

$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial u}{\partial t} \Rightarrow A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ EV = (1, 1, 0) Parabolic

$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} - \frac{\partial^2 u}{\partial t^2} = 0 \Rightarrow A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ EV = (1, 1, -1) Hyperbolic

Note: For 2 independent variables

$$A \frac{\partial^2 u}{\partial x^2} + 2B \frac{\partial^2 u}{\partial x \partial y} + C \frac{\partial^2 u}{\partial y^2} = f\left(u, \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}\right)$$

If $B^2 - AC \begin{cases} = 0 & \text{Parabolic} \\ > 0 & \text{Hyperbolic} \\ < 0 & \text{Elliptical} \end{cases}$

* OPERATORS :-

- $AX = B$ $\rightarrow$ vector $\rightarrow$ for discrete domain
 $\begin{matrix} \text{operator} \\ \text{Matrix} \\ (m \times n) \end{matrix} \quad \begin{matrix} \text{vector} \\ (n \times 1) \end{matrix} \quad \begin{matrix} \text{vector} \\ (m \times 1) \end{matrix} \quad :- \quad \mathbb{R}^n \xrightarrow{\text{operator}} \mathbb{R}^m$

- For continuous domain:- $L u = u$
 eg) $\frac{d}{dx} (f) = g$; $\int_a^b f dx = g$

* Linear Operator :- $L(\alpha u + \beta v) = \alpha L(u) + \beta L(v)$
 $\Rightarrow L$ is a linear operator

If an operator is linear, we can use the principle of linear superposition to make calculations easier.

Eg) $\frac{\partial^2 u}{\partial x^2} = x$; At $x=0$, $u=1$
 $x=1$, $u=2$ 3 sources of non Homogeneity

$$u = u_1 + u_2 + u_3$$

$$u_1: \frac{\partial^2 u_1}{\partial x^2} = x$$

$$x=0 \int u=0$$

$$u_2: \frac{\partial^2 u_2}{\partial x^2} = 0$$

$$x=0 \quad u_2=1$$

$$u_3: \frac{\partial^2 u_3}{\partial x^2} = 0$$

$$x=1 \quad u_3=0$$

On solving :- $u_1 = \frac{x^3 - x}{6}$ $u_2 = 1 - x$ $u_3 = 2x$

$$\Rightarrow u = \frac{x^3 + 5x + 6}{6}$$

- For PDEs:-

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$$

$$t=0$$

$$u = u_{01}$$

$$x=0$$

$$u = u_{02}$$

$$x=1$$

$$u = u_{03}$$

$$u = u_1 + u_2 + u_3$$

$$u_1: \frac{\partial^2 u_1}{\partial x^2} = \frac{\partial u_1}{\partial t}$$

$$\left. \begin{array}{l} t=0 \quad u=u_{01} \\ x=0 \quad u=0 \\ x=1 \quad u=0 \end{array} \right\}$$

well posed problem.
 Separation of variables

ill posed

$$U_2: \frac{\partial^2 U_2}{\partial x^2} = \frac{\partial U_2}{\partial t} \quad \begin{matrix} t=0 \\ x=0 \\ x=1 \end{matrix} \quad \begin{matrix} U_2=0 \\ U_2=U_0 \\ U_2=0 \end{matrix} \quad \left. \vphantom{\frac{\partial^2 U_2}{\partial x^2}} \right\} \text{direct separation of variables type solution}$$

$$U_3: \frac{\partial^2 U_3}{\partial x^2} = \frac{\partial U_3}{\partial t} \quad \begin{matrix} t=0 \Rightarrow U_3=0 \\ x=0 \Rightarrow U_3=0 \\ x=1 \Rightarrow U_3=U_0 \end{matrix} \quad \left. \vphantom{\frac{\partial^2 U_3}{\partial x^2}} \right\} \text{Ill posed problem. Can't be solved directly.}$$

Now,

$$U_2(x,t) = U_2^s(x) + U_2^t(x,t) \rightarrow \text{separation of variable}$$

$$\frac{\partial}{\partial t} (U_2^s + U_2^t) = \frac{\partial^2}{\partial x^2} (U_2^s + U_2^t)$$

$$\Rightarrow \frac{\partial U_2^t}{\partial t} = \frac{\partial^2 U_2^s}{\partial x^2} + \frac{\partial^2 U_2^t}{\partial x^2}$$

We can write this as:

$$U_2^s(x): \frac{\partial^2 U_2^s}{\partial x^2} = 0 \quad (1)$$

$$U_2^t(x,t): \frac{\partial U_2^t}{\partial t} = \frac{\partial^2 U_2^t}{\partial x^2} \quad (2)$$

Note: well posed problem:

- ① I.C. is Non Homogeneous
- ② B.C.s are Homogeneous

$$\Rightarrow \text{At } x=0 \quad U_2 = U_0$$

$$\Rightarrow U_2^s + U_2^t = U_0$$

$$\text{We can write this as: At } x=0 \quad U_2^s = U_0, U_2^t = 0$$

$$\text{At } x=1 \quad U_2 = 0$$

$$\Rightarrow U_2^s + U_2^t = 0 \Rightarrow x=1, \quad U_2^s = 0 \quad U_2^t = 0$$

$$\text{At } t=0 \quad U_2 = 0$$

$$\Rightarrow U_2^s + U_2^t = 0 \Rightarrow U_2^t(x, t=0) = -U_2^s$$

time independent

Now, from ①, we know $u_2^s = C_1 x + C_2$
 Applying B.C. we get $u_2^s = u_{02}(1-x)$

$\Rightarrow u_2^t(t=0) = u_{02}(x-1) \rightarrow$ Now separation of variable to solve

This has now become a well posed problem. ✓

For u_3 ,

$$u_3 : u_3^s(x) + u_3^t(x,t) \Rightarrow u_3^s : \frac{d^2 u_3^s}{dx^2} = 0 \quad \begin{matrix} x=0 & u_3^s=0 \\ x=1 & u_3^s=u_{03} \end{matrix}$$

self sufficient problem

$$u_3^t : \frac{\partial u_3^t}{\partial t} = \frac{\partial^2 u_3^t}{\partial x^2} \quad \begin{matrix} x=0 & u_3^t=0 \\ x=1 & u_3^t=0 \\ t=0 & u_3^t = -u_3^s(x) \end{matrix}$$

Eg 2)

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$$

$$\text{At } t=0 \quad u = u_0$$

$$x=0 \quad \frac{\partial u}{\partial x} = +q_0 \quad (\text{N.B.C})$$

$$x=1 \quad u = u_{02}$$

$$u = u_1 + u_2 + u_3$$

$$\text{For } u_1 : \frac{\partial u_1}{\partial t} = \frac{\partial^2 u_1}{\partial x^2}$$

well posed

$$\text{At } t=0 \quad u_1 = u_{01}$$

$$x=0 \quad \frac{\partial u_1}{\partial x} = 0$$

$$x=1 \quad u_1 = 0$$

$$\text{For } u_2 : \frac{\partial u_2}{\partial t} = \frac{\partial^2 u_2}{\partial x^2}$$

well posed

$$\text{At } t=0 \quad u_2 = 0$$

$$x=0 \quad \frac{\partial u_2}{\partial x} = q_0$$

$$x=1 \quad u_2 = 0$$

$$\text{For } u_3 : \frac{\partial u_3}{\partial t} = \frac{\partial^2 u_3}{\partial x^2}$$

$$\text{At } t=0 \quad u_3 = 0$$

$$x=0 \quad \frac{\partial u_3}{\partial x} = 0$$

$$x=1 \quad u_3 = u_{02}$$

$$U_2(x, t) = U_2^s(x) + U_2^t(x, t)$$

For U_2^s : $\frac{d^2 U_2^s}{dx^2} = 0$ At $x=0$ $\frac{dU_2^s}{dx} = 0$ $x=1$ $U_2^s = 0$ } self sufficient

For U_2^t : $\frac{\partial^2 U_2^t}{\partial x^2} = \frac{\partial U_2^t}{\partial t}$ At $x=0$ $\frac{\partial U_2^t}{\partial x} = 0$ $x=1$ $U_2^t = 0$
 $t > 0$ $U_2^t = -u_2^s(x)$ } well posed

Eg 3

$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ At $t=0$ $u = u_0$
 $x=0 \Rightarrow \frac{\partial u}{\partial x} + \gamma u = \beta$
 $x=1 \Rightarrow u = u_02$

$$u = u_1 + u_2 + u_3$$

u_1 : $\frac{\partial u_1}{\partial t} = \frac{\partial^2 u_1}{\partial x^2} \rightarrow t=0 \Rightarrow u_1 = u_01$ $x=0 \Rightarrow \frac{\partial u_1}{\partial x} + \gamma u_1 = 0$
 $x=1 \Rightarrow u_1 = 0$ } well posed

u_2 : $\frac{\partial u_2}{\partial t} = \frac{\partial^2 u_2}{\partial x^2} \rightarrow t=0$ $u_2 = 0$ $x=0 \Rightarrow \frac{\partial u_2}{\partial x} + \gamma u_2 = \beta$
 $x=1 \Rightarrow u_2 = 0$ } Ill posed

u_3 : $\frac{\partial u_3}{\partial t} = \frac{\partial^2 u_3}{\partial x^2} \rightarrow t=0$ $u_3 = 0$, $x=0 \Rightarrow \frac{\partial u_3}{\partial x} + \gamma u_3 = 0$
 $x=1 \Rightarrow u_3 = u_02$

$$\text{Let } u_2(x, t) = u_2^s(x) + u_2^t(x, t)$$

$U_2^s = \frac{d^2 U_2^s}{dx^2} = 0$ } self sufficient At $x=0$ $\frac{dU_2^s}{dx} + \gamma U_2^s = \beta$

$$x=1$$

$$v_2^s = 0$$

$$v_2^t: \frac{\partial^2 v_2^t}{\partial x^2} = \frac{\partial v_2^t}{\partial t}$$

well posed

$$\text{At } x=0$$

$$\frac{\partial v_2^t}{\partial x} + q v_2^t = 0$$

$$x=1$$

$$v_2^t = 0$$

$$t=0$$

$$v_2^t = -v_2^s(x)$$

* SPECIAL FUNCTIONS

① In discrete domain:- $AX = \lambda X \rightarrow$ Eigenvalue Problem
We are getting Eigenvalue and Eigenvector

② Continuous domain:- $Lu + \lambda \eta u = 0$
Here, η is a weight factor which is 1 for rectangular co-ordinates.

$Lu = -\lambda u$: Mandatory B.C.s need to be homogeneous
Advantage: We can get corresponding Eigenvalue to Eigen functions

$$\text{D. B.C :- } \Rightarrow \frac{d^2 y}{dx^2} + \lambda y = 0 \text{ at } x=0 \text{ \& } x=1 \Rightarrow y=0$$

Case 1 :-

$$\lambda = 0 \Rightarrow \frac{d^2 y}{dx^2} = 0 \Rightarrow y = C_1 x + C_2$$

$$[C_1 = 0, C_2 = 0 \Rightarrow y = 0] \rightarrow \text{trivial sol}^n$$

Case 2 :-

$$\lambda = -\alpha^2 = -q^2 \Rightarrow \frac{d^2 y}{dx^2} - \alpha^2 y = 0 \Rightarrow y = C_1 e^{\alpha x} + C_2 e^{-\alpha x}$$

Applying B.C. we get: $C_1 + C_2 = 0$

$$C_1 e + \frac{C_2}{e} = 0$$

we get

$$C_1 = C_2 = 0$$

trivial solⁿ

Case-1

$$\lambda = +ve = \alpha^2 \Rightarrow \frac{d^2 y}{dx^2} + \alpha^2 y = 0 \Rightarrow y = C_1 \sin(\alpha x) + C_2 \cos(\alpha x)$$

$$x=0, y=0 \Rightarrow C_2 = 0 \quad \therefore y = C_1 \sin(\alpha x)$$

$$x=1, y=0 \Rightarrow C_1 \sin(\alpha) = 0 \Rightarrow \alpha_n = n\pi \rightarrow \text{Eigenvalues}$$

$$y_n = C_n \sin(C_n \pi x) : \text{Eigenfunction}$$

$$\Rightarrow y = \sum_{n=1}^{\infty} y_n = \sum_{n=1}^{\infty} C_n \sin(C_n \pi x)$$

$$b) \frac{d^2 y}{dx^2} + \lambda y = 0 \quad \text{At } x=0 \quad \frac{dy}{dx} = 0 \quad x=1 \quad y=0$$

Case 1:

$$\lambda = 0 \Rightarrow y = C_1 x + C_2 \Rightarrow C_1 = 0 \quad C_2 = 0 \quad \therefore y = 0 \rightarrow \text{Trivial}$$

Case 2:

$$\lambda = -ve = -\alpha^2 \Rightarrow y = C_1 e^{\alpha x} + C_2 e^{-\alpha x}$$

$$\text{From B.C.s} \rightarrow C_1 e + C_2 e^{-1} = 0 \quad \alpha C_1 - \alpha C_2 = 0$$

$$\Rightarrow C_1 = C_2 = 0 \rightarrow \text{Trivial}$$

Case 3:

$$\lambda = +ve = \alpha^2$$

$$y = C_1 \sin(\alpha x) + C_2 \cos(\alpha x)$$

$$\text{From B.C.} \rightarrow C_1 \sin \alpha + C_2 \cos \alpha = 0$$

$$\& \quad C_1 \alpha = 0$$

$$\Rightarrow y = C_2 \cos(\alpha x) \quad \& \quad \alpha = (2n+1)\pi/2 \rightarrow \text{Eigenvalue}$$

$$\Rightarrow y_n = C_n \cos(C_n \pi x) \rightarrow \text{Eigenfunction}$$

$$y = \sum_{n=1}^{\infty} C_n \cos(C_n \pi x)$$

$$c) \quad \frac{d^2 y}{dx^2} + \lambda y = 0 \quad x=0, y=0 \quad x=1 \quad \frac{dy}{dx} + \beta y = 0$$

Case-1

$$\lambda = 0 \Rightarrow y = C_1 x + C_2 \Rightarrow C_1 = 0, C_2 = 0 \Rightarrow y = 0 \rightarrow \text{Trivial}$$

Case 2

$$\lambda = -ve = -\alpha^2 \Rightarrow y = C_1 e^{\alpha x} + C_2 e^{-\alpha x}$$

$$\text{From B.C.} \Rightarrow C_1 + C_2 = 0$$

$$C_1 (\alpha + \beta) e^{\alpha} + C_2 (\alpha - \beta) e^{-\alpha} = 0 \quad \therefore C_1 = C_2 = 0 \rightarrow \text{Trivial}$$

Case-3

$$\lambda = +ve = \alpha^2 \Rightarrow y = C_1 \cos(\alpha x) + C_2 \sin(\alpha x) = 0$$

$$\text{From B.C.} \quad C_1 = 0 \quad \text{and} \quad C_2 (\alpha \cos(\alpha) + \beta C_2 \sin(\alpha)) = 0$$

$$C_2 [\alpha + \beta \tan(\alpha)] = 0 \Rightarrow \alpha_n + \beta \tan(\alpha_n) = 0$$

The roots of this equation will give us infinite eigenvalues. We can use N.R. method to get the roots. All the roots will be in Arithmetic Progression with 3 adding to each roots. We can start with 0, after finding the next root, we use the previous root + 3 to get the new root.

For non standard B.C., transformation is done such that we convert B.C. to standard.

$$\text{eg ①} \quad \frac{d^2 y}{dx^2} + \alpha^2 y = 0 \quad x=0, y=0 \quad \& \quad x=1 \quad \frac{dy}{dx} = 0$$

$$x' = 1-x \quad \text{At } x'=0, \frac{dy}{dx'} = 0 \quad \& \quad x'=1, y=0$$

$$\Rightarrow y = \sum C_n \cos(\alpha_n x) = \sum C_n \cos(\alpha_n (1-x)) \quad \alpha_n = (2n-1)\pi/2$$

$$\textcircled{2} \quad \frac{d^2 y}{dx^2} + \alpha^2 y = 0 \quad \text{At } x=1, y=0$$

$$x=0 \quad \frac{dy}{dx} + \beta y = 0$$

$$X=1-x \Rightarrow x=0 \quad y=0 \quad \text{and } x=1 \quad \frac{dy}{dx} + \beta y = 0$$

This will give solution

$$y_n - \beta \tan(\alpha_n) = 0$$

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* ODEs WITH SPECIAL PROPERTIES

① $d^2y/dx^2 + \lambda y = 0$

$$By = 0$$

→ Legendre's Eqn

② $(1-x^2)y'' - 2xy' + n(n+1)y = 0$

→ Application in Spherical co-ordinates

$$-1 \leq x \leq 1$$

$$\text{at } x = \pm 1, y = 0$$

$$\Rightarrow y(x) = C_1 P_n(x) + C_2 Q_n(x)$$

Legendre's
Polynomial of
order n

Legendre's
Function of order n

- Properties of Q_n : Q_n is unbounded at $x = \pm 1$
 $\Downarrow$
 $C_2 = 0$ $y(x) = C_1 P_n(x)$

- Properties of Legendre:
Polynomial

$$P_n(1) = 1$$

$$P_n(-1) = (-1)^n$$

$$\int_{-1}^1 P_n^2(x) dx = \frac{2}{2n+1}$$

$$\int_{-1}^1 P_n(x) P_m(x) dx = 0$$

→ Orthogonal property for $m \neq n$
Inner product

$$(3) \quad x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + x^2 y = 0$$

↳ 0th order Bessel Equation

↳ cylindrical co-ordinate system

Solⁿ: $y(x) = C_1 J_0(x) + C_2 Y_0(x)$

0th order Bessel fⁿ of first kind

0th order Bessel fⁿ of second kind

$$(4) \quad x^2 y'' + x y' + (x^2 - n^2) y = 0$$

↳ Generalised nth order Bessel eqⁿ

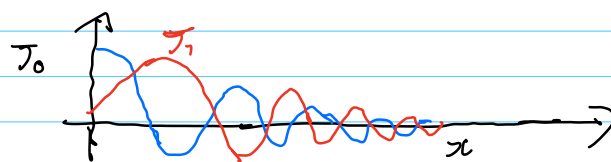
Solⁿ $y(x) = C_1 J_n(x) + C_2 Y_n(x)$

nth order Bessel fⁿ of first kind

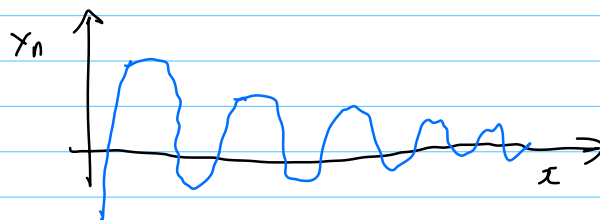
second kind

Properties: ① J_n is bounded at $x=0$

② Y_n is unbounded at $x=0$



At $x=0$ $J_n \rightarrow$ finite/bounded



At $x=0$ $Y_n \rightarrow$ infinite/bounded

$$(5) \quad x^2 y'' + a x y' + b y = 0$$

↳ Euler's Equation

↳ Spherical Co-ordinate System

Soln: $y \sim x^{\eta}$
substitute

$$\eta(\eta-1) + a\eta + b = 0 \rightarrow \text{Characteristic Equation}$$

$\swarrow \quad \searrow$
 $\eta_1 \quad \eta_2$

$$\Rightarrow y(x) = C_1 x^{\eta_1} + C_2 x^{\eta_2}$$

* EVALUATION OF ADJOINT OPERATOR

For a discrete domain

$$AX = B$$

$$\Rightarrow X = A^{-1}B \rightarrow \text{To evaluate}$$

For a continuous domain

$$LU = f$$

$$\Rightarrow U = L^{-1}f$$

Operator

↳ To evaluate. It's called **ADJOINT OPERATOR (L^*)**

For $\alpha \leq x \leq \beta$

$$LU = f \rightarrow \beta u = 0$$

For simplicity, we consider a one dimensional operator.

$$L u = a_0(x) \frac{d^2 u}{dx^2} + a_1(x) \frac{du}{dx} + a_2(x) u$$

$$\Rightarrow L = a_0(x) \frac{d^2}{dx^2} + a_1(x) \frac{d}{dx} + a_2(x)$$

$$\Rightarrow L^* = \langle \psi, Lu \rangle = \int_{\alpha}^{\beta} \psi L u dx$$

$$= \int_{\alpha}^{\beta} v (a_0 v'' + a_1 v' + a_2 v) dx$$

$$= \int_{\alpha}^{\beta} v a_0 v'' dx + \int_{\alpha}^{\beta} v a_1 v' dx + \int_{\alpha}^{\beta} v a_2 v dx$$

By parts

$$= v a_0 v' \Big|_{\alpha}^{\beta} - \int_{\alpha}^{\beta} (v' a_0) v' dx + v a_1 v \Big|_{\alpha}^{\beta} - \int_{\alpha}^{\beta} (v' a_1) v dx$$

$$= [v a_0 v' + v a_1 v]_{\alpha}^{\beta} - \int_{\alpha}^{\beta} (v' a_0 + v a_0') v' dx - \int_{\alpha}^{\beta} (v' a_1 + v a_1') v dx + \int_{\alpha}^{\beta} v a_2 v dx$$

By parts

$$= [v a_0 v' + v a_1 v]_{\alpha}^{\beta} - [v' a_0 + v a_0'] v \Big|_{\alpha}^{\beta} + \int_{\alpha}^{\beta} (v' a_0 + v a_0') v dx - \int_{\alpha}^{\beta} (v' a_1 + v a_1') v dx + \int_{\alpha}^{\beta} v a_2 v dx$$

$$= [v a_0 v' + v a_1 v - v' a_0 v - v a_0' v]_{\alpha}^{\beta} + \int_{\alpha}^{\beta} [a_0 v' + a_0 v'' + a_0' v' + v a_0'' - v' a_1 - v a_1' + v a_2] v dx$$

Bilinear
Concomittant

$$= J(v, v) + \int_{\alpha}^{\beta} [a_0 v'' + 2 a_0' v' - a_1 v' + (a_0'' - a_1' + a_2) v] v dx$$

$$= J(v, v) + \int_{\alpha}^{\beta} [\underline{a_0} v'' + (\underline{2a_0' - a_1}) v' + (\underline{a_0'' - a_1' + a_2}) v] v dx$$

Hence

$$L^* = a_0(x) \frac{d^2}{dx^2} + (2a_0' - a_1) \frac{d}{dx} + (a_0'' - a_1' + a_2)$$

$Lu = f \rightarrow$ Problem

$L^*u = f \rightarrow$ Adjoint Problem

$B^*u = ??$

For $J(u, v) = 0$, to select the BCs of v

- If $L = L^*$ & $B = B^*$
 $\rightarrow$ self adjoint problem

Eg1) $Lu = d^2y/dx^2$
 $L = d^2/dx^2$

DRC: at $x=0, 1$ $y=0$
 $L^* = ?$ $B^* = ?$

$$\langle v, Lu \rangle = \int_0^1 v \frac{d^2u}{dx^2} dx$$

$$= v \frac{du}{dx} \Big|_0^1 - \int_0^1 \frac{dv}{dx} \cdot \frac{du}{dx} dx$$

$$= v \frac{du}{dx} \Big|_0^1 - \frac{dv}{dx} \cdot u \Big|_0^1 + \int_0^1 \frac{d^2v}{dx^2} u dx$$

$$= \left[v \frac{du}{dx} - u \frac{dv}{dx} \right]_0^1 + \int_0^1 \frac{d^2v}{dx^2} u dx$$

$J(u, v)$

$$= \left[v(1) \frac{du}{dx} \Big|_1 - v(0) \frac{du}{dx} \Big|_0 - u(1) \frac{dv}{dx} \Big|_1 + u(0) \frac{dv}{dx} \Big|_0 \right] + \int_0^1 u L^* v dx$$

$\langle u, L^* v \rangle$

Now

selecting a BC

at $x=0$ $v=0$

$x=1$ $v=0$

to make $J(u, v) = 0$

$$= \langle u, L^* v \rangle$$

$$\Rightarrow L = L^* \quad , \quad B = B^* \rightarrow \text{Self Adjoint Problem}$$

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$$\text{Ex 2)} \quad L = \frac{d^2}{dx^2} \quad \frac{d^2 u}{dx^2} = f(x)$$

$$\text{B.C. at } x=0 \quad du/dx = 0$$

$$x=1 \quad u=0$$

$$L^* = ? \quad B^* = ?$$

$$\langle L u, v \rangle$$

$$= \int_0^1 v \frac{d^2 u}{dx^2} dx$$

$$= v \frac{du}{dx} \Big|_0^1 - \int_0^1 \frac{dv}{dx} \frac{du}{dx} dx$$

$$= v \frac{du}{dx} \Big|_0^1 - \frac{dv}{dx} u \Big|_0^1 + \int_0^1 \frac{d^2 v}{dx^2} u dx$$

$$= \left[v \frac{du}{dx} - u \frac{dv}{dx} \right]_0^1 + \int_0^1 v L^* u dx$$

$$= \langle u, L^* v \rangle + J(u, v)$$

$$\Rightarrow L^* = \frac{d^2}{dx^2} = L$$

$$J(u, v) = \left[v \frac{du}{dx} - u \frac{dv}{dx} \right]_0^1$$

$$= v \frac{du}{dx} \Big|_1 - \cancel{u \frac{dv}{dx} \Big|_1} - v \frac{du}{dx} \Big|_0 - \cancel{u \frac{dv}{dx} \Big|_0}$$

$$= u(1) \left. \frac{dv}{dx} \right|_1 + v(0) \left. \frac{du}{dx} \right|_0 \quad \text{has to be 0 for } J(u,v) \text{ to vanish}$$

$$\Rightarrow \left. \begin{array}{l} \text{at } x=0 \\ x=1 \end{array} \right\} \begin{array}{l} \frac{du}{dx} = 0 \\ v = 0 \end{array} \quad B^*$$

$$L = L^* \quad B = B^* \Rightarrow \text{Self Adjoint}$$

Ex 3 Same question diff B.C.

$$B \left\{ \begin{array}{l} \text{at } x=0 \\ x=1 \end{array} \right. \begin{array}{l} v = 0 \\ \frac{dv}{dx} + \beta v = 0 \end{array}$$

$$L^* = ? \quad B^* = ?$$

Clearly, following above derivation $L^* = L$

$$J(u,v) = u(1) \left. \frac{dv}{dx} \right|_1 - v(1) \left. \frac{du}{dx} \right|_1 - v(0) \left. \frac{du}{dx} \right|_0 + \cancel{v(0) \left. \frac{dv}{dx} \right|_0}$$

$$= -u(1) \beta v(1) - v(1) \left. \frac{du}{dx} \right|_1 - v(0) \left. \frac{du}{dx} \right|_{x=0}$$

$$= -v(1) \left[\frac{du}{dx} + \beta u \right]_{x=1} - v(0) \left. \frac{du}{dx} \right|_{x=0}$$

$$\Rightarrow B^* \left[\begin{array}{l} x=0 \\ x=1 \end{array} \right. \begin{array}{l} v = 0 \\ \frac{du}{dx} + \beta u = 0 \end{array} \Rightarrow B = B^* \quad \downarrow \quad \text{Self Adjoint}$$

$$* \quad a_0(x) \frac{d^2 u}{dx^2} + a_1(x) \frac{du}{dx} + a_2(x) u + \lambda a_3(x) u = 0$$

$$\text{at } x=0 \quad \alpha_1 \frac{du}{dx} + \beta_1 u = 0$$

$$\swarrow \quad \searrow \quad x=1 \quad \alpha_2 \frac{du}{dx} + \beta_2 u = 0$$

Homogeneous
B.C.

This equation is of the form $Lu = -\lambda a_3 u$
 $= -\lambda' u$

⇒ GENERALISED EIGENVALUE PROBLEM
 OR
 STURM - LOUVILLE PROBLEM

Is the SL operator self adjoint?

$$\downarrow$$

$$L = a_0(x) \frac{d^2}{dx^2} + a_1(x) \frac{d}{dx} + a_2(x) \quad - (1)$$

This can be recast as:

$$\frac{d}{dx} \left[p(x) \frac{du}{dx} \right] + q(x) u + \lambda r(x) u = 0 \quad - (2)$$

$$\text{If } p(x) = \exp \left[\int \frac{a_1(x)}{a_0(x)} dx \right]$$

$$q(x) = \frac{a_2(x)}{a_0(x)} p$$

$$r(x) = \frac{a_3(x)}{a_0(x)} p$$

Proof: $p = \exp \left[\int \frac{a_1(x)}{a_0(x)} dx \right]$

$$\ln p = \int \frac{a_1(x)}{a_0(x)} dx \Rightarrow \frac{1}{p} \frac{dp}{dx} = \frac{a_1(x)}{a_0(x)}$$

$$\Rightarrow \frac{dp}{dx} = \frac{a_1 C(x)}{a_0(x)} p$$

Substituting in (2)

$$p \frac{d^2 u}{dx^2} + \frac{dp}{dx} \frac{du}{dx} + q u + \lambda r u = 0$$

$$p \frac{d^2 u}{dx^2} + \frac{a_1}{a_0} p \frac{du}{dx} + \frac{a_2}{a_0} p u + \frac{\lambda a_3}{a_0} p u = 0$$

$$\Rightarrow a_0 \frac{d^2 u}{dx^2} + a_1 \frac{du}{dx} + a_2 u + \lambda a_3 u = 0$$

Identical

Since (1) and (2) are identical $\Rightarrow$ their operators must also be identical

$$\frac{d}{dx} \left[p \frac{du}{dx} \right] + q u + \lambda r u = 0$$

$$\Rightarrow L = \frac{d}{dx} \left[p \frac{d}{dx} \right] + q = p \frac{d^2}{dx^2} + p' \frac{d}{dx} + q$$

Clearly, comparing coefft we get

$$\begin{aligned} p &= a_0 & p' &= a_1 \\ q &= a_2 \end{aligned}$$

From last class notes

$$\text{If } L u = a_0 u'' + a_1 u' + a_2 u$$

Then

$$L^* u = a_0 u'' + (2a_0' - a_1) u' + (a_0'' - a_1' + a_2) u$$

$$L^* = L \rightarrow \text{Self Adjoint}$$

$$\begin{aligned}
 \Rightarrow \mathcal{L}^* &= a_0 \frac{d^2}{dx^2} + (2a_0' - a_1) \frac{d}{dx} + (a_0'' - a_1' + a_2) \\
 &= p \frac{d^2}{dx^2} + (2p' - p') \frac{d}{dx} + (p'' - p'' + q) \\
 &= p \frac{d^2}{dx^2} + p' \frac{d}{dx} + q
 \end{aligned}$$

$$\begin{array}{ll}
 \text{B.C} & x=a \quad \alpha_1 u' + \alpha_2 u = 0 \\
 & x=b \quad \beta_1 u' + \beta_2 u = 0
 \end{array}$$

we know

$$\begin{aligned}
 J(u, v) &= \left[u \alpha_0 v' - u' \alpha_0 v - u \alpha_0' v + u' \alpha_0 v \right]_a^b \\
 &= \left[u \alpha_0 v' - u' \alpha_0 v \right]_a^b
 \end{aligned}$$

$$\begin{aligned}
 &= u(b) \alpha_0(b) v'(b) - u'(b) \alpha_0(b) v(b) - u(a) \alpha_0(a) v'(a) \\
 &\quad + u'(a) \alpha_0(a) v(a)
 \end{aligned}$$

$$\begin{aligned}
 &= u \alpha_0 \left[-\frac{\beta_2}{\beta_1} v \right] \Big|_b - u' \alpha_0 v \Big|_b - u \alpha_0 \left[-\frac{\alpha_2}{\alpha_1} v \right] \Big|_a \\
 &\quad + u' \alpha_0 v \Big|_a
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow J(u, v) &= -\alpha_0 v \left[\frac{\beta_2}{\beta_1} u + u' \right] \Big|_b + \alpha_0 v \left[\frac{\alpha_2}{\alpha_1} u + u' \right] \Big|_a \\
 &= \frac{\alpha_0 v}{\alpha_1} \left[\alpha_2 u + \alpha_1 u' \right] \Big|_a - \frac{\alpha_0 v}{\beta_1} \left[\beta_2 u + \beta_1 u' \right] \Big|_b
 \end{aligned}$$

Hence to make it zero

$$\text{At } x=a \quad \alpha_1 \frac{du}{dx} + \alpha_2 u = 0$$

$$x=b \quad \beta_1 \frac{du}{dx} + \beta_2 u = 0$$

$B^* = B$
SELF ADJOINT

* Thm 1:- S-L problem is a standard eigenvalue problem in continuous domain, having infinite no. of eigenvalues and eigenfunctions.

* Definition:- If ϕ_m and ϕ_n are two functions, orthogonal wrt weight function $\delta(x)$

$\int \phi_m(x) \phi_n(x) \delta(x) dx = 0$
 for $m \neq n$

* Thm 2:- If λ_m and λ_n are two distinct eigenvalues then corresponding eigenfunctions y_m and y_n are orthogonal functions wrt weight function δ .

$$\begin{aligned}
 Ly &= -\lambda \delta(x) y \\
 \Rightarrow Ly_m &= -\lambda_m \delta(x) y_m \quad \text{① } (y_m \rightarrow \text{eigen f}^n) \\
 Ly_n &= -\lambda_n \delta(x) y_n \quad \text{②}
 \end{aligned}$$

$$\langle \text{①}, y_n \rangle = \langle \text{②}, y_m \rangle$$

$$\begin{aligned}
 \langle y_n, Ly_m \rangle &= \langle y_m, Ly_n \rangle \\
 &= \int \lambda_m \delta(x) y_m y_n dx \\
 &\quad + \int \lambda_n \delta(x) y_n y_m dx
 \end{aligned}$$

$$\begin{aligned}
 \langle L^* y_n, y_m \rangle &= \int \delta(x) y_m y_n dx = \langle y_m, Ly_n \rangle \\
 \langle y_m, Ly_n \rangle &= \langle Ly_m, y_n \rangle = (\lambda_n - \lambda_m) \int y_m y_n \delta dx \\
 0 &= (\lambda_n - \lambda_m) \int y_m y_n \delta dx
 \end{aligned}$$

$\Rightarrow$ orthogonal

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* SOLUTIONS OF PDES

↓

Parabolic	}	Rectangular coord	}	2D
Elliptic		Cylindrical coord		3D
Hyperbolic		Spherical coord		4D

$$\textcircled{1} \quad \frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} \quad \begin{array}{l} \text{at } t=0 \quad u = u_0 \\ \text{at } x=0 \\ \quad \quad \quad =1 \end{array} \left. \vphantom{\frac{\partial u}{\partial t}} \right\} u=0$$

Separation of variables

$$u(x, t) = T(t)X(x)$$

$$X \frac{dT}{dt} = T \frac{d^2 X}{dx^2} \Rightarrow \frac{1}{T} \frac{dT}{dt} = \frac{1}{X} \frac{d^2 X}{dx^2} = \text{const} = 0, \alpha^2, -\alpha^2$$

We'll get trivial solⁿ for 0 and α^2

$$\Rightarrow \frac{d^2 X}{dx^2} + \alpha^2 X = 0 \quad \text{at } x=0 \left. \vphantom{\frac{d^2 X}{dx^2}} \right\} X=0$$

Solving, and applying B.C. we get

$$\alpha_n = n\pi \quad n=1, 2, \dots, \infty$$

$$X_n(x) = C_1 \sin(C_n \pi x)$$

$$\frac{1}{T_n} \frac{dT_n}{dt} = -n^2 \pi^2 \Rightarrow T_n(t) = C_2 \exp(-n^2 \pi^2 t)$$

$$u_n = C_n \sin(C_n \pi x) \exp(-n^2 \pi^2 t)$$

$$u = \sum_{n=1}^{\infty} u_n = \sum_{n=1}^{\infty} C_n \sin(C_n \pi x) \exp(-n^2 \pi^2 t)$$

$$\text{At } t=0, \quad u = u_0$$

$$\Rightarrow u_0 = \sum_n C_n \sin(C_n \pi x)$$

We know, thanks to orthogonality

$$\int_0^1 \sin(n\pi x) \sin(m\pi x) dx = 0 \quad \forall m \neq n$$

$$\begin{aligned} \Rightarrow U_0 \int_0^1 \sin(m\pi x) dx &= \int_0^1 \sum_n C_n \sin(n\pi x) \sin(m\pi x) dx \\ &= C_m \int_0^1 \sin^2(m\pi x) dx \quad (\text{all other terms } 0) \end{aligned}$$

$$\Rightarrow U_0 \int_0^1 \sin(m\pi x) dx = \frac{C_m}{2}$$

↪ change m to running index n

$$2 U_0 \left[\frac{1 - \cos n\pi}{n\pi} \right] = \frac{C_n}{2}$$

$$\Rightarrow U_n = 2 U_0 \sum_{n=1}^{\infty} \left(\frac{1 - \cos n\pi}{n\pi} \right) \exp(-n^2 \pi^2 t) \sin(n\pi x)$$

$$\begin{aligned} \textcircled{2} \quad \frac{\partial u}{\partial t} &= \frac{\partial^2 u}{\partial x^2} & t=0 & \quad u = U_0 \\ & & x=0 & \quad \frac{\partial u}{\partial x} = 0 \\ & & x=1 & \quad u = 0 \end{aligned}$$

Once again, following same procedure we get

$$\begin{aligned} \frac{d^2 X}{dx^2} + \alpha^2 X &= 0 & x=0 & \quad dX/dx = 0 \\ & & x=1 & \quad X = 0 \end{aligned}$$

on solving

$$\alpha_n = (2n-1) \frac{\pi}{2} \quad n = 1, 2, \dots, \infty$$

$$X_n = C_1 \cos(\alpha_n x)$$

$$T_n(t) = C_2 \exp(-\alpha_n^2 t)$$

$$\Rightarrow v_n = C_n \cos(q_n x) \exp(-q_n^2 t)$$

$$\Rightarrow v = \sum_{n=1}^{\infty} v_n = \sum_n C_n \cos(q_n x) \exp(-q_n^2 t)$$

$$\text{At } t=0, v=v_0$$

$$v_0 = \sum_n C_n \cos(q_n x)$$

$$v_0 \int_0^1 \cos(q_m x) dx = \int_0^1 \sum_n C_n \cos(q_n x) \cos(q_m x) dx$$

$\downarrow$ changing m to n
orthogonality

$$= C_n \int_0^1 \cos^2(q_n x) dx$$

$$v_0 \frac{\sin(q_n)}{q_n} = \frac{C_n}{2} \Rightarrow C_n = \frac{2v_0 \sin(q_n)}{q_n}$$

$$\Rightarrow v(x, t) = 2v_0 \sum_{n=1}^{\infty} \frac{\sin(q_n)}{q_n} \cos(q_n x) \exp(-q_n^2 t)$$

$$\textcircled{3} \quad \frac{\partial v}{\partial t} = \frac{\partial^2 v}{\partial x^2} \quad \begin{array}{ll} \text{at } t=0 & v=v_0 \\ x=0 & v=0 \\ x=1 & \frac{\partial v}{\partial x} + \beta v = 0 \end{array}$$

once again, solving similarly we get:

$$\frac{d^2 X}{dx^2} + q^2 X = 0$$

$$\text{at } x=0 \quad X=0$$

$$x=1 \quad \frac{dX}{dx} + \beta X = 0$$

$$q_n + \beta \tan q_n = 0$$

Roots of the eqⁿ are eigenvalues.
 $X_n = C_1 \sin(q_n x)$

$$T(t) = C_2 \exp(-q_n^2 t)$$

$$u(x, t) = \sum_{n=1}^{\infty} u_n = \sum_{n=1}^{\infty} C_n \exp(-\alpha_n^2 t) \sin(\alpha_n x)$$

$$\text{At } t=0, u = u_0$$

$$u_0 = \sum C_n \sin(\alpha_n x)$$

$$u_0 \int_0^1 \sin(\alpha_m x) dx = \int_0^1 \sum_{n=1}^{\infty} C_n \sin(\alpha_m x) \sin(\alpha_n x) dx$$

$\downarrow$
 $m \rightarrow n$

$$u_0 \left[\frac{1 - \cos(\alpha_n)}{\alpha_n} \right] = C_n \int_0^1 \sin^2(\alpha_n x) dx$$

$$= \frac{C_n}{2} \int_0^1 (1 - \cos(2\alpha_n \pi x)) dx$$

$$u_0 \left[\frac{1 - \cos \alpha_n}{\alpha_n} \right] = \frac{C_n}{2} \left[1 - \frac{\sin(2\alpha_n)}{2\alpha_n} \right]$$

$$= \frac{C_n}{2} \left[1 - \frac{1}{\alpha_n} \frac{\tan \alpha_n}{1 + \tan^2 \alpha_n} \right]$$

$$= \frac{C_n}{2} \left[1 + \frac{1}{\alpha_n} \frac{\alpha_n/\beta}{1 + \alpha_n^2/\beta^2} \right]$$

$$= \frac{C_n}{2} \left[1 + \frac{\beta}{\beta^2 + \alpha_n^2} \right]$$

$$\Rightarrow C_n = 2u_0 \left[\frac{1 - \cos \alpha_n}{\alpha_n} \right] \left[\frac{\alpha_n^2 + \beta^2}{\alpha_n^2 + \beta^2 + \beta} \right]$$

$$u(x, t) = \sum_{n=1}^{\infty} C_n \sin(\alpha_n x) \exp(-\alpha_n^2 t)$$

Q) 1-Dimensional Transient Heat Transfer across the slab

$$\rho C_p \frac{\partial T}{\partial t} = k \frac{\partial^2 T}{\partial x^2} \rightarrow \text{Model Equation}$$

$$\begin{array}{lll} \text{At } x=0 & T=T_1 & t=0 \quad T=T_0 \\ x=L & T=T_2 & \end{array}$$

To make it non-dimensional, $\Theta = \frac{T-T_1}{T_1}$ $x^* = x/L$

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} \quad \alpha = \frac{k}{\rho c_p} \rightarrow \text{m}^2/\text{s} \rightarrow \text{Thermal Diffusivity}$$

$$Q(t) = L^2/\alpha$$

$\downarrow$
order of time

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$$\frac{\partial \Theta}{\partial t} = \frac{\alpha}{L^2} \frac{\partial^2 \Theta}{\partial x^{*2}} \quad \Rightarrow \quad \frac{L^2}{\alpha} \frac{\partial \Theta}{\partial t} = \frac{\partial^2 \Theta}{\partial x^{*2}}$$

$$\text{Let } \tau = \frac{\alpha t}{L^2} \Rightarrow \frac{\partial \Theta}{\partial \tau} = \frac{\partial^2 \Theta}{\partial x^{*2}}$$

$$\text{at } \tau=0 \quad \Theta = \frac{T_0 - T_1}{T_1} = \Theta_0$$

$$\text{at } x^*=0 \quad \Theta = 0$$

$$x^*=1 \quad \Theta = \frac{T_2 - T_1}{T_1} = \Theta_{10} \rightarrow \text{Ill posed}$$

To convert it into well posed,

$$\Theta = \Theta_1 + \Theta_2$$

$$\Theta_1: \quad \frac{\partial \Theta_1}{\partial \tau} = \frac{\partial^2 \Theta_1}{\partial x^{*2}}$$

$$\text{at } \tau=0 \quad \Theta_1 = \Theta_0$$

$$x^*=0 \quad \Theta_1 = 0$$

$$x^*=1 \quad \Theta_1 = 0$$

$$\text{We know } \Theta_1 = \Theta_0 \sum_{n=1}^{\infty} \left(\frac{1 - \cos n\pi}{n\pi} \right) e^{-n^2 \pi^2 \tau} \sin(n\pi x^*)$$

$$\Theta_2: \frac{\partial \Theta_2}{\partial z} = \frac{\partial^2 \Theta_2}{\partial x^{*2}} \quad \text{at } z=0 \quad \Theta_2=0$$

$$x^*=0 \quad \Theta_2=0$$

$$x^*=1 \quad \Theta_2=\Theta_{10}$$

→ Ill posed

$$\Theta_2(x^*, z) = \Theta_2^s(x^*) + \Theta_2^t(x^*, z)$$

$$\Theta_2^s: \frac{\partial^2 \Theta_2^s}{\partial x^{*2}} = 0 \Rightarrow \Theta_2^s(x^*) = C_1 x^* + C_2$$

$$\text{At } x^*=0 \quad \Theta_2^s=0$$

$$x^*=1 \quad \Theta_2^s=\Theta_{10}$$

$$\Rightarrow \Theta_2^{cs} = \Theta_{10} x^*$$

$$\Theta_2^t: \frac{\partial^2 \Theta_2^t}{\partial x^{*2}} = \frac{\partial \Theta_2^t}{\partial z}$$

$$x^*=0 \quad \Theta_2^t=0$$

$$x^*=1 \quad \Theta_2^t=0$$

$$t=0 \quad \Theta_2^t = -\Theta_2^s(x^*) \rightarrow \text{well posed}$$

$$\text{On solving } \Theta_2^t = \sum C_n e^{-n^2 \pi^2 z} \sin(n \pi x^*)$$

$$\text{at } z=0, \Theta_2^t = -\Theta_2^s(x^*)$$

$$\sum C_n \sin(n \pi x^*) = -\Theta_{10} x^*$$

Multiplying both sides by $\sin m \pi x^* dx^*$ and integrating

$$-\Theta_{10} \int_0^1 x^* \sin(n \pi x^*) dx^* = \frac{C_n}{2}$$

$$\Rightarrow C_n = -2\Theta_{10} \int_0^1 x^* \sin(n \pi x^*) dx^*$$

$$= -2\Theta_{10} \left[\frac{-x^* \cos(n \pi x^*)}{n \pi} \Big|_0^1 + \int_0^1 \frac{\cos(n \pi x^*)}{n \pi} dx^* \right]$$

$$= -2\Theta_{10} \left[-\frac{\cos n \pi}{n \pi} \right] = \frac{2\Theta_{10} \cos(n \pi)}{n \pi}$$

$$\Theta = \Theta_1 + \Theta_2^S + \Theta_2^T$$

Q2 > $\frac{\partial T}{\partial t} = \gamma \frac{\partial^2 T}{\partial x^2}$ at $t=0$ $T = T_0$
 at $x=0$ $-k \frac{\partial T}{\partial x} = q_0$
 at $x=1$ $T = T_1$

Now, let $\Theta = \frac{T - T_1}{T_1}$ $x^* = x/L$ $\tau = \frac{t\gamma}{L^2}$

$$\frac{\partial \Theta}{\partial \tau} = \frac{\partial^2 \Theta}{\partial x^{*2}}$$

$\tau = 0$ $\Theta = \frac{T_0 - T_1}{T_1} = \Theta_0$

$x^* = 0$ $-\frac{k T_1}{L} \frac{\partial \Theta}{\partial x^*} = q_0$

$$\Rightarrow \frac{\partial \Theta}{\partial x^*} = -\frac{q_0 L}{k T_1} = q_0^*$$

$x^* = 1 \Rightarrow \Theta = 0$

$$\Theta = \Theta_1 + \Theta_2$$

$\Theta_1: \frac{\partial \Theta_1}{\partial \tau} = \frac{\partial^2 \Theta_1}{\partial x^{*2}}$

$\tau = 0 \Rightarrow \Theta_1 = \Theta_0$

$x^* = 0 \Rightarrow \frac{\partial \Theta_1}{\partial x^*} = 0$

$x^* = 1$ $\Theta_1 = 0$

} well posed

$\alpha_n = (2n-1)\pi/2$ $n=1, \dots, \infty$

$$\Theta_1(x^*, \tau) = 2\Theta_0 \sum_{n=1}^{\infty} \frac{\sin \alpha_n}{\alpha_n} \exp(-\alpha_n^2 \tau) \cos(\alpha_n x^*)$$

$\Theta_2: \frac{\partial \Theta_2}{\partial \tau} = \frac{\partial^2 \Theta_2}{\partial x^{*2}}$

at $\tau = 0$, $\Theta_2 = 0$

$x^* = 0$ $\frac{\partial \Theta_2}{\partial x^*} = q_0^*$

$x^* = 1$ $\Theta_2 = 0$

well posed

$$\Theta_2(x^*, z) = \Theta_2^s(x^*) + \Theta_2^t(x^*, z)$$

$$\Theta_2^s : \frac{d^2 \Theta_2^s}{dx^{*2}} = 0 \quad \text{at } x^* = 0 \quad \frac{d\Theta_2^s}{dx^*} = q_0$$

$$\quad \quad \quad \text{at } x^* = 1 \quad \Theta_2^s = 0$$

$$\Theta_2^s = C_1 x^* + C_2$$

$$\text{Solving } \Theta_2^s = q_0^* (x^* - 1)$$

$$\Theta_2^t : \frac{d^2 \Theta_2^t}{dz^2} = \frac{d\Theta_2^t}{dz}$$

$$z = 0 \quad \Theta_2^t = -\Theta_2^s(x^*)$$

$$x^* = 0 \quad \frac{d\Theta_2^t}{dz} = 0$$

$$x^* = 1 \quad \Theta_2^t = 0$$

$$\eta_n = (n-1)\pi/2 \quad n = 1, 2, \dots, \infty$$

$$\Theta_2^t = \sum_{n=0}^{\infty} C_n e^{-\eta_n^2 z} \cos(\eta_n x^*)$$

$$\Rightarrow -\Theta_2^s(x^*) = q_0(1-x^*) = \sum_{n=1}^{\infty} C_n \cos(\eta_n x^*)$$

$$\Rightarrow C_n = 2q_0 \int_0^1 (1-x^*) \cos(\eta_n x^*) dx$$

Can be solved further

$$\Theta = \Theta_1 + \Theta_2^s + \Theta_2^t$$

$$Q3) \quad \frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} \quad \text{at } t=0 \quad T=T_0$$

$$x=0 \quad T=T_1$$

$$x=L \quad -k \frac{\partial T}{\partial x} = h(T-T_\infty)$$

$$\text{Let } \Theta = \frac{T-T_\infty}{T_1-T_\infty} \quad x^* = x/L \quad \tau = \frac{t\alpha}{L^2}$$

$$\Rightarrow \frac{\partial \Theta}{\partial \tau} = \frac{\partial^2 \Theta}{\partial x^{*2}}$$

$$\text{at } \tau=0 \quad \Theta = \frac{T_0-T_\infty}{T_1-T_\infty} = \Theta_0$$

$$x^*=0 \quad \Theta=1$$

$$x^*=1 \quad -\frac{k(T_1-T_\infty)}{L} \frac{\partial \Theta}{\partial x^*} = h(T_1-T_\infty)\Theta$$

$$\Rightarrow \frac{\partial \Theta}{\partial x^*} + \frac{hL}{k} \Theta = 0$$

$$\Rightarrow \frac{\partial \Theta}{\partial x^*} + Bi \Theta = 0 \quad \rightarrow \text{Biot Number}$$

$$\text{Now, let } \Theta = \Theta_1 + \Theta_2$$

$$\frac{\partial \Theta_1}{\partial \tau} = \frac{\partial^2 \Theta_1}{\partial x^{*2}}$$

$$\text{at } \tau=0 \quad \Theta_1 = \Theta_0$$

$$x^*=0 \quad \Theta_1=0$$

$$x^*=1 \quad \frac{\partial \Theta_1}{\partial x^*} + Bi \Theta_1 = 0$$

$$\alpha_n + Bi \tan(\alpha_n) = 0 \quad \rightarrow \text{roots}$$

$$C_n = 2\Theta_0 \left[\frac{1-\cos \alpha_n}{\alpha_n} \right] \left[\frac{\alpha_n^2 + Bi^2}{\alpha_n^2 + Bi^2 + Bi} \right]$$

$$\Theta_1 = \sum_{n=1}^{\infty} C_n \sin(\alpha_n x^*) \exp(-\alpha_n^2 \tau)$$

$$\Theta_z : \frac{\partial \Theta_z}{\partial z} = \frac{\partial^2 \Theta_z}{\partial x^{*2}}$$

$$\text{at } z=0 \quad \Theta_z = 0$$

$$x^* = 0 \quad \Theta_z = 1$$

$$x^* = 1 \quad \frac{\partial \Theta_z}{\partial x^*} + \beta_i \Theta_z = 0$$

$$\Theta_z = \Theta_z^f(x^*) + \Theta_z^b(x^*, z)$$

$$\Theta_z^f : \frac{\partial^2 \Theta_z^f}{\partial x^{*2}} = 0 \quad \Theta_z^f = C_1 x^* + C_2$$

$$x^* = 0 \quad \Theta_z^f = 1$$

$$x^* = 1 \quad \frac{d\Theta_z}{dx^*} + \beta_i \Theta_z = 0$$

$$\Rightarrow C_2 = 1$$

$$C_1 + \beta_i (C_1 + 1) = 0$$

$$C_1 (1 + \beta_i) = -\beta_i \Rightarrow C_1 = \frac{-\beta_i}{1 + \beta_i}$$

$$\Theta_z^f = \frac{-\beta_i x^*}{1 + \beta_i} + 1$$

$$\Theta_z^b : \frac{\partial^2 \Theta_z^b}{\partial x^{*2}} = \frac{\partial \Theta_z^b}{\partial z}$$

$$z=0 \quad \Theta_z^b = -\Theta_z^f$$

$$x^*=0 \quad \Theta_z^b = 0$$

$$x^*=1 \quad \frac{\partial \Theta_z}{\partial x^*} + \beta_i \Theta_z = 0$$

$$\Theta_z^b = \sum_{n=0} C_n \sin(C_n x^*) \exp(-C_n^2 z)$$

$$\frac{\beta_i x^*}{1 + \beta_i} - 1 = \sum C_n \sin(C_n x^*)$$

$$\int_0^1 \left(\frac{Bix^2}{1+Bi} - 1 \right) \sin(q_n x^2) dx^2 = c_n \int_0^1 \sin^2(q_n x) dx$$